

Plan4ARI

Planning for Artificial Robotic Intelligence and Beyond

Integrated architecture for trustworthy industrial autonomy

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Motivation and Vision

- **Acronym:** Plan4ARI.
- **Title:** Planning for Artificial Robotic Intelligence and Beyond.
- **Duration:** 48 months.
- **Area:** Advanced Manufacturing.
- **PI:** Nicola Pedrocchi.
- **Host Institution:** COMAU.
- **Research Organization:** CNR.

Strategic goal

Develop an **Artificial Robotic Intelligence (ARI)** stack that makes industrial robots easier to program, safer to deploy, and more adaptive in dynamic human-robot environments.

Why Plan4ARI?

- Industrial robots are powerful but still constrained by **specialized programming**, lengthy engineering cycles, and limited adaptation to changing environments.
- The proposal argues that recent progress in **large language models, deliberative AI, deep sensing, adaptive motion planning, and neural control** can now be systematically integrated.
- The central thesis is that cognitive capability must be distributed across **all controller layers**: user interaction, task planning, motion planning, and real-time execution.
- The result is not a generic AGI claim, but a practical **Industry 5.0 architecture** for trustworthy robotic autonomy.

AGI perspective

The proposal frames AGI as the long-term ambition of systems able to understand, learn, and reason across tasks.

Industrial constraint

Manufacturing does not need unconstrained general intelligence; it needs **robust, safe, and verifiable intelligence** embedded in robot workflows.

ARI perspective

Plan4ARI defines **Artificial Robotic Intelligence** as the coordinated integration of:

- natural language programming,
- task allocation and scheduling,
- scene understanding and safe motion planning,
- adaptive real-time motion control.

Four Levels of Robotic Intelligence

- ① **Programming intelligence:** the user describes the task instead of coding it.
- ② **Task intelligence:** the robot allocates and schedules actions across humans and other robots in the scene.
- ③ **Motion intelligence:** the robot plans safe movement under contextual constraints.
- ④ **Execution intelligence:** the controller adapts online to scene evolution and disturbances.

Implication

Plan4ARI is built as a layered cognitive stack, not as a single AI module bolted onto a conventional robot.

- A **high-payload mobile manipulator** performs welding operations inside a large boat or ship structure.
- The environment is **unstructured, harmful, and dynamically evolving**; references are weak and many operations are defined on site.
- A human operator supervises multiple robots and interacts with the system in **natural language**.
- The system must support virtual preview, error refinement, safe execution, and notification when unexpected conditions arise.

Why this matters

The scenario is not science fiction: the proposal argues that most building blocks already exist, but must be turned into a **new integrated controller architecture**.

Industrial Pain Points Addressed

Layer	Current limitation	Plan4ARI response
Programming	Process experts depend on robot specialists and low-code flows remain complex and suboptimal	Natural-language-based generation of verified flowcharts
Task planning	Plans are hard to verify, brittle under uncertainty, and weakly optimized	KB-driven PDDL, formal checks, BT/STNU and MCTS optimization
Motion planning	Traditional planners struggle with dynamic scenes, human presence and run-time adaptation	Sensor fusion, informed sampling, safety-aware replanning, MPC micro-interpolation
Control	Industrial controllers are difficult to tune and cope poorly with uncertainty and elasticity	Prescribed-performance neural control and knowledge reuse
HRI adoption	Trust, workload and acceptance limit deployment in real factories	Subjective and objective trustworthiness assessment integrated in the loop

COMAU

Industrial automation and robotics leader, providing the host platform, controller access, laboratories, and targeted demonstrators.

CNR-STIIMA

Manufacturing systems, industrial robotics, controls, mechatronics, digital methods, and technology transfer.

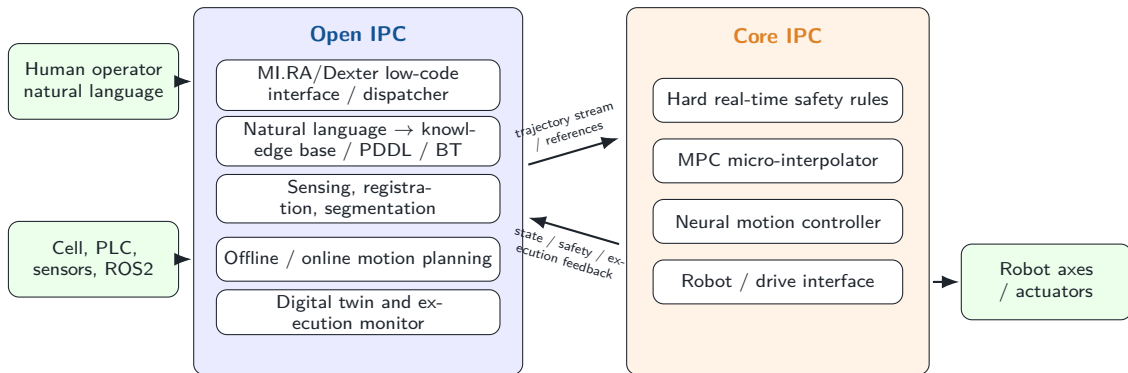
Strategic fit

The project combines **industrial integration capability** with **scientific depth** across planning, robotics, control, sensing and human factors.



Architecture and Technical Concept

Conceptual Architecture: Two-IPC Design

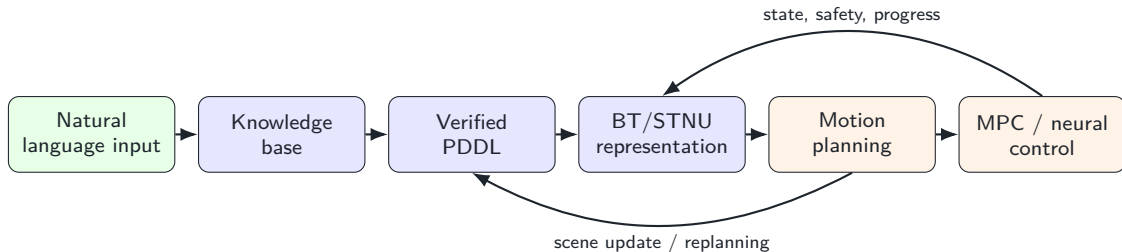


Interpretation

The Open IPC hosts cognition, world modelling and planning. The Core IPC preserves deterministic execution, safety enforcement and high-rate control.

- The **Open IPC** acts as the bridge between the robot controller and the external world:user interaction, PLCs, sensors, ROS2, digital twin, and asynchronous planning.
- The **Core IPC** concentrates hard-real-time functions:safety, micro-interpolation, and high-rate motion control.
- This split is essential because the proposal separates **non-real-time cognitive functions** from **deterministic control execution**.
- The architecture therefore enables innovation while preserving safety-critical behaviour.

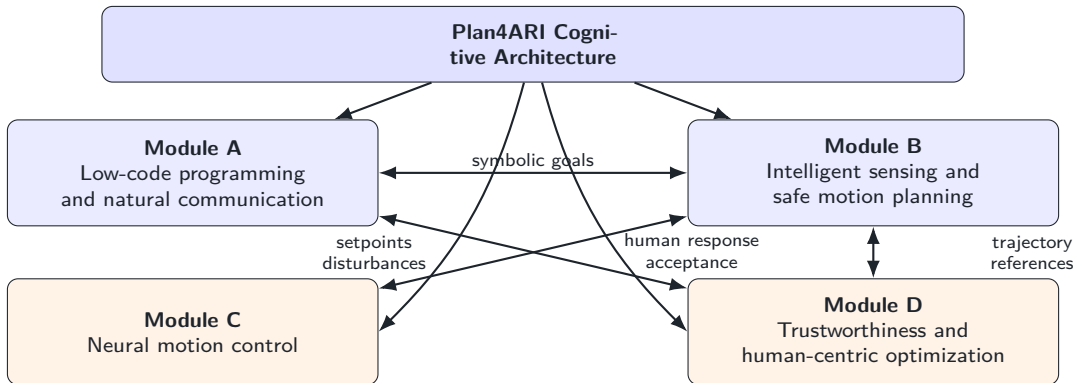
Technical Workflow from User to Robot



Key point

Plan4ARI links all abstraction levels: symbolic planning, geometric planning, and control are coupled through explicit feedback and execution monitoring.

Project Modules



Meaning

The proposal does not define independent work packages only; it defines a **functional decomposition of cognition** where outputs from one module directly condition the others.

- MI.RA/Dexter acts as **dispatcher** and **execution monitor** in deliberative-AI terms.
- It supports a model-driven, low-code representation where atomic software functionalities are composed into process pipelines.
- It hosts scene registration, semantic segmentation, and motion planning modules, and interfaces with ROS2.
- It virtualizes operations for the user, enabling preview, validation and monitoring.
- It is also the natural insertion point for future AI copilots for process-flow generation.

- Runs the **real-time motion controller** and a high-rate motion planner or micro-interpolator.
- Enforces safety rules and deterministic execution constraints.
- Receives streamed references from Open IPC and turns them into real-time setpoints.
- Hosts **Model Predictive Control** for micro-interpolation.
- Hosts the **neural controller** with prescribed-performance design, enabling faster tuning and better tracking in changing conditions.

- The proposal explicitly states that Plan4ARI will be developed in compliance with the **EU AI Act**, emphasizing safety, transparency, data protection and fundamental rights.
- At operational level, human tracking, sensing and decision support are framed with **GDPR-style mitigation principles**: anonymization, online-only processing where possible, and ethics checks.
- At control level, collaboration is tied to **ISO/TS 15066**-style safety reasoning and to trustworthiness metrics developed in Module D.

Takeaway

This is not just a performance-driven robotics project; it is a **trustworthy AI robotics architecture**.

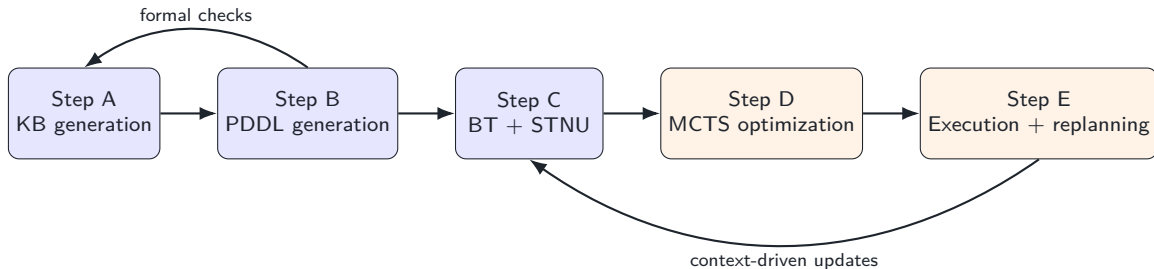


Scientific Module A: Low-Code Programming and Natural Communication

- The objective is to let a **process expert** program a robot through natural language rather than through detailed coding or block-level manual orchestration.
- The module builds on COMAU's MI.RA/Dexter and addresses the limitations of current low-code tools: library knowledge, sequence design burden, and weak optimization.
- The ambition is not only translation but **verified and optimized program generation**.

- Recent work such as **ProgPrompt** and **Text2Motion** shows that LLMs can translate language into robot plans or TAMP structures.
- **PDDLStream** demonstrates the value of combining symbolic and black-box components for planning.
- **Behavior Trees from temporal plans** show a promising route toward execution in open-world robotic applications.
- The gap highlighted in the proposal is that these approaches remain insufficient for industrial deployment because they are not deterministic enough, do not natively manage temporal uncertainty, and often lack hard verification.

Five-Step Methodology



Interpretation

The methodology turns free-form language into a constrained planning pipeline that can be validated, optimized in a digital twin, and adapted at execution time.

Step A - Knowledge-Base Generation

- Generative AI is first used to create a **Knowledge Base (KB)** from a natural-language description of a class of processes.
- The KB captures **types, predicates and actions**, i.e., the formal domain description that planning needs.
- The process engineer validates the KB architecture; this is a **human-in-the-loop domain foundation** rather than blind end-to-end generation.
- Target evolution: from **TRL2 to TRL4** according to the proposal.

Step B - PDDL from Natural Language

- Once the domain is represented in the KB, natural-language task descriptions are converted into **PDDL descriptors** coherent with that domain.
- The proposal prefers this two-step strategy over direct end-to-end PDDL generation because a prior KB improves consistency and reduces hallucinated actions.
- Multi-agent generative AI and a **formal verification stage** iteratively refine the PDDL until it matches the stored domain definition.
- Target evolution: **TRL4 to TRL6** for the PDDL generation / verification chain.

Step C - Behavior Trees with Temporal Uncertainty

- The proposal converts verified PDDL into a **Behavior Tree** enriched by time-variance modelling.
- The enabling formalism is **STNU** (Simple Temporal Networks with Uncertainty), extending approaches based on standard STNs.
- The purpose is to capture duration variability without waiting for a perfect off-the-shelf temporal planner dealing with uncertainty.
- This is a core scientific novelty because it links symbolic planning to robust execution under variability.

Step D / E - Optimization, Execution and Replanning

- The BT representation is optimized inside a **digital twin** using **Monte Carlo Tree Search** to identify the best policy under temporal variance.
- The result is then executed by a software module that runs the optimal BT and supports **online replanning**.
- The proposal distinguishes two runtime modes:
 - an **anytime planning** module that continuously improves the plan while execution proceeds;
 - a **heuristic restart** strategy for sudden, disruptive scene changes where feasibility matters more than optimality.

Activity 2 (WP2) - Tasks, Timing and Outputs

Task	Timing	Focus
T2.1	M7-M12	Update the scientific plan according to the fast evolution of generative AI
T2.2	M13-M24	Lightweight LLM-based PDDL2 generation for robotic tasks
T2.3	M18-M24	PDDL2 to Behavior Tree generation with time-variance modelling
T2.4	M22-M30	Task and Motion Planning optimization in a digital twin environment
T2.5	M28-M36	Optimal BT execution and online replanning
T2.6	M28-M36	Integration inside MI.RA/Dexter low-code interface platform
T2.7	M13-M36	Continuous integration and versioning
T2.8	M7-M12	Benchmarking and evaluation setup

Module A - Expected Results, KPIs and TRL

Indicator	KPI / target	TRL target
Programming time reduction	66% reduction	-
Programming by non-pretrained worker	Demonstrated	-
KB correctness	10 contexts evaluated	TRL2 → TRL4
PDDL correctness	100+ contexts	TRL2 → TRL4
BT generation correctness	100+ contexts	TRL4 → TRL6
Optimal task planning evaluation	100+ contexts	TRL2 → TRL4
Fleet management	3 mobile manipulators or 4 industrial arms	-



Scientific Module B: Intelligent Sensing and Safe Motion Planning

- Module B couples **scene awareness** and **safe motion planning**. In the proposal, these are explicitly treated as inseparable.
- The challenge is collaborative robotics in dynamic workspaces where humans, mobile manipulators, and objects share the scene.
- The module spans sensing fusion, registration, semantic segmentation, symbolic-to-geometric grounding, offline planning, online replanning, and high-rate motion adaptation.

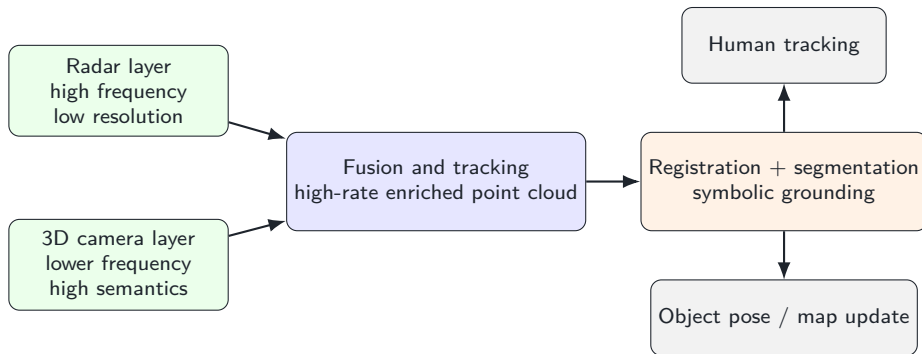
Why Sensing and Planning Are Intertwined

- Task planning depends on knowing **which objects are present**, where they are, and how humans move in the workspace.
- Reactive motion adaptation only works if scene updates are timely and geometrically meaningful.
- Plan4ARI therefore aims at a **fully autonomous toolchain**: symbolic task descriptions should automatically become geometric and kinematic commands.

Implication

Scene understanding is not only for perception; it is a prerequisite for **safe and productive autonomy**.

Sensor-Fusion Concept



Proposed advantage

Radar provides fast updates, while 3D vision provides accurate geometry and semantics; fusion should support safer and more responsive planning.

- Shared workspaces require more than collision-free motion: speed limitation and stops are triggered by safety modules implementing collaborative operations such as **SSM** and **PFL**.
- The proposal distinguishes **proactive** planning (avoiding expected interference beforehand) and **reactive** planning (modifying the path online).
- Existing methods have complementary weaknesses: proactive approaches degrade badly out of distribution; reactive approaches can lead to frequent stops and reduced efficiency.
- Plan4ARI explicitly aims to bridge them through **safety-aware replanning** coupled to human-aware costs.

Offline / asynchronous level

- global path computation,
- anytime replanning,
- digital-twin virtualization,
- Open IPC deployment.

Runtime / synchronous level

- scene-aware path repair,
- velocity modulation,
- visual-servo style synchronous planning,
- Core IPC MPC micro-interpolation.

Key design choice

Planning is distributed across computation units according to latency and determinism requirements.

- The module builds on sampling-based planning, especially **RRT-like** methods and informed sets.
- The proposal introduces a **mixed strategy**: global admissible informed sampling for asymptotic optimality, and local sampling around the current solution for fast improvement.
- The trade-off between exploration and exploitation is regulated adaptively using online rewards.
- A graph connecting pre-computed paths to the same goal supports efficient **multi-path replanning** when the current path is blocked.

- A central proposal idea is that a path is not good enough if it is only collision-free; it must also reduce the intervention of the safety system.
- The planner therefore includes a **human-aware cost** based on ISO/TS 15066-inspired logic and uncertain human-state estimation.
- This extends prior work by the PI on safety-aware time-optimal motion planning and directly targets improved human-robot cooperation efficiency.

Expected benefit

Fewer unnecessary slowdowns and stops, hence better productivity in collaborative scenarios.

- The proposal uses **Model Predictive Control** not only as feedback control, but as a trajectory-generation and micro-interpolation engine.
- MPC predicts future robot states over a finite horizon, incorporates limits on joints, velocity, acceleration and workspace, and produces smooth feasible references.
- In Plan4ARI, MPC is meant to improve tracking under load, interface naturally with sensing (including visual servoing), and exploit residual degrees of freedom to reduce jerk or motor load.

Activity 3 (Sensing for Context Awareness)

Task	Timing	Focus
T3.1	M1-M6	Update scientific plan for segmentation and pose estimation methodologies
T3.2	M7-M9	Market study and integration of radar sensors for scene monitoring
T3.3	M10-M16	Radar-camera fusion and tracking using multi-object methods
T3.4	M17-M24	Benchmarking and integration of semantic segmentation and 3D pose estimation methods

Expected output

A sensing stack packaged as MI.RA/Dexter functional blocks and able to support tracking, registration and symbolic grounding.

Activity 4 (Local and Global Autonomous Motion Planning)

Task	Timing	Focus
T4.1	M1-M6	Update on motion-planning state of the art
T4.2	M4-M24	Informed sample-based optimal planning in dynamic environments
T4.3	M4-M24	MPC for adaptive micro-interpolation
T4.4	M25-M30	Integration in MI.RA/Dexter
T4.5	M31-M36	Integration of MPC inside COMAU Core IPC
T4.6	M4-M36	Continuous integration
T4.7	M1-M6	Benchmarking and evaluation design

Module B - Expected Results, KPIs and TRL

Indicator	KPI / target	TRL target
Motion-plan generation	Fully autonomous joint and Cartesian plans	-
Interaction execution time	30% reduction vs. state-of-the-art planners	-
Computation time benchmark	Mean over 100+ contexts	TRL5 → TRL7
Scene-update / registration accuracy	About 20% better than standard 3D-vision registration	TRL4 → TRL6
Tracking performance under dynamics load	About 50% improvement	TRL5 → TRL7
Safe velocity adaptation	Demonstrated under varying working conditions	-



Scientific Module C: Neural Motion Control

- In the proposal, the inner layer of ARI is the **high-rate real-time motion controller**.
- Conventional industrial control relies on accurate models, full-state observers and careful tuning; this is powerful but expensive to deploy and difficult in the presence of elasticity and uncertainty.
- Plan4ARI seeks a new generation of **adaptive neural controllers** that can run on industrial PCs while preserving robustness and safety.

- The proposal reviews several observer families: **UKF**, **sliding-mode observers**, and **high-gain observers**. Each has strengths, but compromises appear in computational burden, noise sensitivity, or robustness.
- Neural control is appealing because it naturally handles nonlinear dynamics, disturbances, and rich sensing inputs.
- The remaining barriers are **computational burden**, **robustness demonstration**, and integration with sensing and industrial real-time constraints.

- The proposal combines **Prescribed Performance Control (PPC)** with adaptive learning, specifically **RBF neural networks**.
- PPC predefines acceptable transient and steady-state behavior; learning then estimates and compensates unknown dynamics while respecting those envelopes.
- The controller is designed to cope with **modelling uncertainty, actuator failures, motion constraints, and joint elasticity**.

- ① Define a dynamic learning scheme for industrial manipulators including lumped elastic effects.
- ② Transform the constrained tracking problem into an unconstrained stabilization variable through an **error transformation**.
- ③ Use filtered errors and adaptive neural control to force tracking errors to a small neighborhood while satisfying prescribed performance.
- ④ Transform the closed-loop dynamics into an LTV structure with small perturbations, enabling accurate convergence of neural weights.
- ⑤ Store the converged weights so that learned knowledge can be **reused** on similar tasks.

Why the Proposed Controller Matters

Claimed advantages

- easy integration of compliance modelling,
- no need for a full-state observer,
- MIMO design without matrix inversion,
- straightforward integrator modelling,
- limited computation time.

Industrial consequence

The controller should reduce tuning effort, shorten deployment time, and improve high-dynamics tracking performance without violating industrial real-time constraints.

Activity 5 (Neural Robust Motion Control)

Task	Timing	Focus
T5.1	M1-M6	Scientific-plan update for control methodologies
T5.2	M4-M30	Design and proof of the neural controller with prescribed-performance constraints
T5.3	M12-M36	Integration in COMAU Open IPC / platform wrapper
T5.4	M4-M36	Continuous integration
T5.5	M1-M6	Benchmarking and evaluation setup

Module C - Expected Results, KPIs and TRL

Indicator	KPI / target	TRL target
Auto-tuning of dynamics/control parameters	Demonstrated	-
Mean tracking error	About 10% reduction	TRL5 → TRL7
Overshoot under high dynamics	About 30% reduction	TRL5 → TRL7
Computation time stress tests	Mean over 200+ contexts	TRL2 → TRL5
Deployment time of a running controller	About 50% reduction vs current COMAU time	TRL4 → TRL6



Scientific Module D: Trustworthiness and Human-Centric Optimization

- The proposal treats trustworthiness not as an afterthought, but as a component of the robotic architecture.
- Adoption barriers are cultural as well as technical: workers must trust hardware, software, safety logic and the quality of robot cooperation.
- The project therefore develops both **subjective** and **objective** measures of human acceptance and interaction quality.

- Design new subject-evaluation forms specific to **AI-powered industrial HRC**.
- Validate them statistically on an initial population and then deploy them on a larger sample.
- Develop objective, ideally non-intrusive metrics based on interaction quality during operation.
- Map objective data and perceived trust to the project **Knowledge Base**, so that human factors can influence decision support and risk assessment.

What Is Measured?

Dimension	Examples in the proposal
Perceived safety	Trust in not being physically damaged; trust in safety software
Task quality	Whether robot and operator effectively accomplish the job
Role / agency	Whether the operator feels leading, equal to, or driven by the cobot
Stress / workload	Mental and physical stress before, during and after shifts
Human-robot relationship	Relevance of the cobot identity, tool vs colleague perception
Dynamic interaction quality	Intention recognition, space negotiation, unexpected breaks in cooperation

Activity 6 (Human-Robot Trustworthiness Evaluation and Assessment)

Task	Timing	Focus
T6.1	M7-M9	Design of the questionnaire and topic refinement with operators
T6.2	M10-M16	Validation campaign with at least 50 workers
T6.3	M17-M24	Statistical refinement and final questionnaire deployment
T6.4	M24-M36	Evaluation campaign and dataset creation with 100+ workers

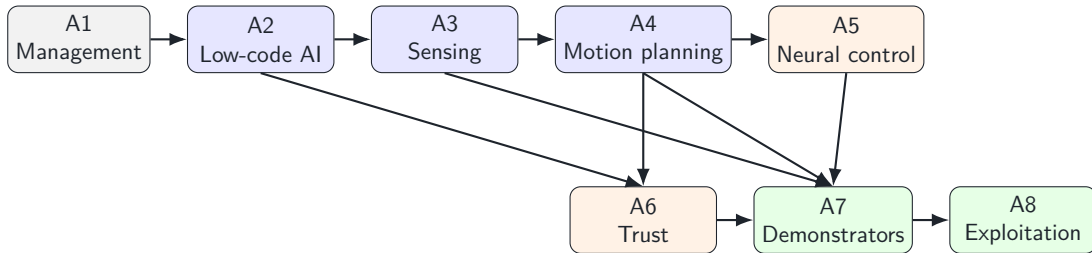
Module D - Expected Results, KPIs and TRL

Indicator	KPI / target	TRL target
Subjective trustworthiness metrics	100+ workers; open dataset statistics	TRL9 (existing generic practice) → tailored TRL5
Objective trustworthiness metrics	10+ consortium workers; open statistics	TRL3 → TRL4
Continuous monitoring	Data-driven and context-aware dynamic measures	-
Integration into control framework	Human-centric optimization and KB update	-



Project Plan, Activities and Management

Project Logic Across Activities



Reading

Management and data governance support all activities. Activities 2–5 deliver the technical stack, Activity 6 validates human acceptance, Activity 7 integrates the final demonstrators, and Activity 8 drives dissemination and exploitation.

Activity 1 - Scientific, Technical and Project Management

- **Objective:** guarantee that technical, financial, administrative and coordination aspects remain aligned with scientific objectives over 48 months.
- **Task 1.1:** general management, project templates, reporting procedures, internal communication, gender-equality attention.
- **Task 1.2:** financial management and audit preparation.
- **Task 1.3:** technical and risk coordination, deliverable quality monitoring, data-management-plan updates, contingency actions.
- **Task 1.4:** research data management with informed consent, anonymization, GDPR compliance and ethics alignment.
- **Task 1.5:** advisory board creation within the first 6 months.

Activity 2 - Low-Code Programming with Generative and Deliberative AI

- **Objective:** reduce programming time and move programming authority from robot experts to process experts.
- Scientific work spans the complete chain: state-of-the-art update, PDDL generation with lightweight LLMs, BT/STNU conversion, MCTS optimization, online replanning, and MI.RA/Dexter integration.
- Continuous integration and early benchmarking are included as dedicated tasks, which is important for industrial software maturity.
- This activity is the main symbolic/planning entry point for the whole architecture.

Activity 3 - Sensing for Context Awareness

- **Objective:** monitor the robot workspace with radar and 3D vision, and provide robust operator tracking and object understanding.
- The work is staged: methodology update, radar acquisition/integration, radar-camera data fusion, benchmarking of segmentation/pose-estimation pipelines.
- The final output is a set of functional sensing blocks embedded in MI.RA/Dexter and connected to motion planning.

Activity 4 - Local and Global Autonomous Motion Planning

- **Objective:** develop planners for teams of manipulators in collaborative dynamic environments.
- The activity combines algorithmic research (informed sample-based planning and MPC micro-interpolation) with direct integration in the COMAU stack.
- The deliverable logic clearly separates algorithm prototyping from the more delicate integration of MPC in the Core IPC.
- Continuous integration and benchmarking are explicit, which should help maintain reproducibility and software quality.

Activity 5 - Neural Robust Motion Control

- **Objective:** push robot low-level control beyond industrial SOTA through neural control algorithms with prescribed-performance constraints.
- Technical work includes modeling of lumped elasticity, controller design and proof, simulation in COMAU-developed environments, and later C++ implementation.
- Integration uses containerized logic and the existing Open IPC / platform interfaces to accelerate industrial deployment while preserving safety engineering discipline.

Activity 6 - Human-Robot Trustworthiness Evaluation and Assessment

- **Objective:** create a tool that helps manufacturers assess workers' engagement and acceptance when operating with AI-powered collaborative robots.
- The activity intentionally follows a staged psychometric process: initial questionnaire design, validation with 50 workers, refinement, and final deployment at 100+ workers.
- This is crucial because the project wants trust metrics that are not generic but **specific to industrial AI-powered HRC**.

Activity 7 - Targeted Demonstrators

- **Objective:** validate the integrated Plan4ARI stack on three sector-specific demonstrators provided by COMAU.
- The demonstrators are not add-ons; they are the project-level integration test for low-code task generation, sensing, planning, control and trustworthiness assessment.
- The three scenarios are: warehouse picking, remanufacturing, and welding in large ships.

Activity 8 - Communication, Dissemination and Exploitation

- **Objective:** maintain communication channels, disseminate scientific and industrial results, and structure exploitation plans and business cases.
- The activity includes website, public materials, digital media, workshops, publications, fairs, and a maintained exploitation culture.
- It also covers market analysis, foreground management, patent surveys, and exploitation planning.

Roles of PI, Research Organization and Host

Actor	Role in the proposal
PI	Scientific coordination, methodology design, continuous technical steering, weekly touch points with teams and COMAU responsables
Research Organization (CNR)	Fundamental and industrial research, methodology definition, prototyping, scientific implementation, benchmarking support
Host Institution (CO-MAU)	Platform access, industrial development teams, sensing and low-level control integration, innovation teams, demonstrator provision and evaluation

Module-Level Resource Allocation Mentioned in the Proposal

Module	Research Org.	Host	Comment
Module A - Low-code	>30%	about 35%	Strong co-development with MI.RA/Dexter teams
Module B - Sensing / motion planning	about 35%	about 35%	Shared effort across perception, planner integration and Core IPC work
Module C - Neural control	about 25%	about 27.5%	Significant industrial integration and digital-twin support
Module D - Trustworthiness	about 10%	about 2.5%	Mainly research-led with innovation-department support

Note

The proposal includes a budget-plan figure and refers to co-financing details in Allegato B; the table above summarizes the explicit percentage statements available in the text.

Deliverables (1/2)

ID	Deliverable	Due
D1.1	Management Report	M12
D1.2	Management Report	M24
D1.3	Management Report	M36
D1.4	Management Report	M48
D2.1	Scientific Plan and First Running Mockup of the Low-Coding Module	M12
D2.2	First Running Time-Variance Aware Planner in MI.RA/Dexter	M24
D2.3	Optimal Task and Motion Planning with online replanning module	M36
D3.1	Scientific Plan and First Running Mockup for the sensing module	M12
D3.2	Sensing Methodologies and First Prototypes Release	M24
D3.3	Sensing Modules integrated in the COMAU IPC	M36
D4.1	Scientific Plan and First Mockup of the Motion Planning Modules	M12
D4.2	Motion Planners Modules Prototypes	M24

Deliverables (2/2)

ID	Deliverable	Due
D4.3	Motion Planners Modules Prototypes integrated in the COMAU IPC	M36
D5.1	Scientific Plan and First Mockup of the Neural Control Architecture	M12
D5.2	First Release of the Neural Controller Prototype	M24
D5.3	Neural Control Architecture integrated in the COMAU IPC	M36
D6.1	Scientific Plan and First Questionnaire Release	M12
D6.2	Second Questionnaire Release	M24
D6.3	Campaign Evaluation Report	M36
D7.1	Demonstrators	-
D8.1	Exploitation, Communication, Dissemination Report	M12
D8.2	Exploitation, Communication, Dissemination Report	M24
D8.3	Exploitation, Communication, Dissemination Report	M36

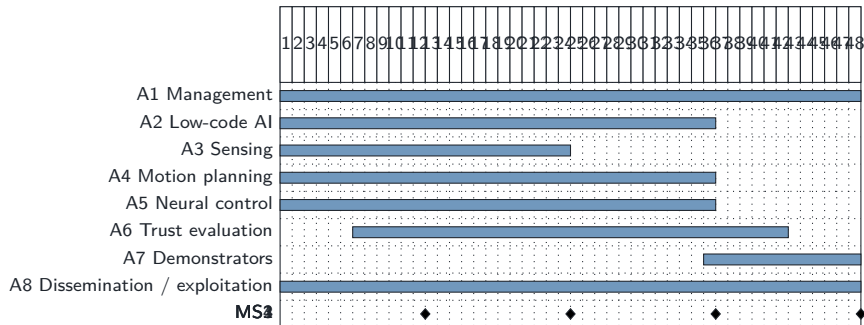
Milestones and Means of Verification

Milestone	Due	Means of verification
MS1	M12	D1.1, D2.1, D3.1, D4.1, D5.1, D6.1, D7.1, D8.1
MS2	M24	D1.2, D2.2, D3.2, D4.2, D5.2, D6.2, D7.2, D8.2
MS3	M36	D1.3, D2.3, D3.3, D4.3, D5.3, D6.3, D7.3, D8.3
MS4	M48	D1.4 and final integrated demonstrator / closure set indicated in the proposal

Observation

Milestones are structured as integration gates coupling management, technical modules, validation and dissemination.

Global GANTT



- **Months 1–12:** scientific-plan updates, benchmarking setup, and first mockups in Modules A-D build the technical baseline for MS1.
- **Months 13–24:** low-code planning chain, sensing fusion, informed planning, and neural-controller design mature to prototype level and define MS2.
- **Months 25–36:** industrial integration in MI.RA/Dexter and COMAU IPC, online replanning, and trust-evaluation campaigns culminate in MS3.
- **Months 36–48:** targeted demonstrators and final exploitation close the project and anchor MS4.



Demonstrators, Impact and Risk

Targeted Demonstrators - Overview

Demo	Use case	Why it matters
D7.1	Warehouse picking of heterogeneous objects	Tests natural-language tasking, object grounding, and collaborative intralogistics
D7.2	Remanufacturing and disassembly of mechanical components	Tests planning, interaction control, force-sensitive behavior and rescheduling
D7.3	Welding of plates in large ship structures with mobile manipulators	Tests operation in harsh, cluttered, safety-critical environments with limited ability to deviate from nominal paths

Demonstrator 1 - Warehouse Picking

- The robot must manipulate a wide variety of objects, including soft or non-standard items, in a dynamic warehouse without rigid layouts.
- The operator may issue commands such as *“Pick the red t-shirt in box A and put it in box B”*.
- The system must translate language into robot programmes, allocate tasks, plan trajectories, continuously monitor the scene, and replan under human presence.
- This demonstrator heavily exercises **KB creation, semantic segmentation, robot localization, and online motion planning**.

Demonstrator 2 - Remanufacturing

- Focuses on disassembly/remanufacturing of mechanical components, where flexible intervention and sustainable manufacturing are key.
- Human operators guide the robot through natural language, while the integrated stack generates optimized task and motion plans.
- The scenario stresses interaction-aware replanning, low-level control performance, and robot-environment force interaction.
- It is the most direct demonstration of the synergy between **task planning, sensing, MPC, and neural control**.

Demonstrator 3 - Ship Welding

- Targets a very harsh industrial domain where welding is still largely manual, harmful and difficult to fence.
- Mobile manipulators must work in huge cluttered structures where operators remain nearby.
- Unlike generic path replanning tasks, welding often forbids significant deviation from the nominal path: the system must stop safely and resume correctly.
- This demonstrator is therefore the most demanding test of **localization, sensing fusion, runtime safety, and human acceptance**.

- Scientifically, the project proposes a **systematic framework for cognitive industrial robotics**, rather than isolated algorithms.
- Industrially, it aims to show that AI can be integrated into standard industrial toolchains to reduce programming effort, improve throughput, reduce errors and improve ergonomics.
- The proposal links its ambition to AI in manufacturing, mobile manipulation and collaborative robotics, stressing European competitiveness and SME accessibility.

- The proposal sets an explicit target of **15% increase in the OECD Job Quality Index** through safer and more ergonomic work environments.
- Expected benefits include reduced accident exposure, lower operator stress, improved information delivery to workers, and better transparency in collaborative workspaces.
- This is one of the strongest distinguishing factors of the project: productivity gains are tied to **human-centric acceptance and workplace quality**.

Requirements, Risks and Mitigation

Risk factor	Proposal concern	Mitigation logic
Market challenger	New entrants may challenge specific AI controllers	Modular open architecture and service-oriented exploitation
Performance	Multi-layer architecture may reduce overall performance	Fine-tuning, solver replacement, suboptimal feasible modes
Scalability	Complex dependencies may hinder reuse	Abstract interfaces and minimized dependencies
Computational burden	Several modules may become heavy	Relax optimality to preserve feasibility
Legal framework	Regulation changes and EU AI Act concerns on biometrics/emotions	GDPR-like anonymization, online-only methods, ethics review
Adoption	Users may reject AI-driven robots	Trust metrics and clear communication strategy

Plan4ARI in one sentence

A 48-month program to turn industrial robots into **trustworthy cognitive agents** by integrating natural language interaction, verified planning, scene-aware motion planning, adaptive real-time control and human-centric trust evaluation.

- Strong industrial anchor through COMAU, with access to real platforms, demonstrators and industrial expertise.
- Strong scientific foundation through CNR-STIIMA, with a clear focus on algorithmic development and integration.
- Clear integration path through eight activities, demonstrators, KPIs, milestones and deliverables.
- High relevance for Industry 5.0, advanced manufacturing and collaborative robotics.